Exact Discrete-Time State-Space and Small-Signal Model of a Triple-Phase-Shift Dual Active Bridge for Bidirectional EV Charging (G2V / V2G)

The triple-phase-shift modulation space is divided into twelve operating modes: six for forward power flow and six for reverse. All modes satisfy the implicit bounds

$$0 \le D_1 \le 1, \qquad 0 \le D_2 \le 1,$$

with $D_3 \ge 0$ for forward power flow and $D_3 \le 0$ for reverse. The mode-dependent constraints are listed in Table 1.

Table 1: TPS operating-mode constraints.

Direction	**Mode**	**Constraints**
Forward	1	$D_1 \ge D_2, \qquad 0 \le D_3 \le D_1 - D_2$
Forward	2	$D_2 \ge D_1, \qquad 1 + D_1 - D_2 \le D_3 \le 1$
Forward	3	$D_2 \le 1 - D_1, \qquad D_1 \le D_3 \le 1 - D_2$
Forward	4	$1 - D_3 \le D_2 \le 1 - D_3 + D_1, \qquad D_1 \le D_3 \le 1$
Forward	5	$D_1 - D_3 \le D_2 \le 1 - D_3, \qquad 0 \le D_3 \le D_1$
Forward	6	$1 - D_2 \le D_1, \qquad 1 - D_2 \le D_3 \le D_1$
Reverse	$1'$	$D_1 \ge D_2, \qquad -1 \le D_3 \le -1 + D_1 - D_2$
Reverse	$2'$	$D_2 \ge D_1, \qquad D_1 - D_2 \le D_3 \le 0$
Reverse	$3'$	$D_2 \le 1 - D_1, \qquad D_1 - 1 \le D_3 \le -D_2$
Reverse	$4'$	$-D_2 \le D_3 \le D_1 - D_2, \qquad D_1 - 1 \le D_3 \le 0$
Reverse	$5'$	$D_1 - 1 - D_2 \le D_3 \le -D_2, \qquad -1 \le D_3 \le D_1 - 1$
Reverse	$6'$	$1 - D_2 \le D_1, \qquad -D_2 \le D_3 \le -1 + D_1$

Each reverse mode follows from its forward counterpart under the substitution

$$D_3^{\text{fwd}} = D_3^{\text{rev}} + 1.$$

Modes 1 and 2 can both charge the battery (G2V) and discharge it (V2G), and so can modes $1'$ and $2'$: in addition to discharging, they can also charge. Modes 3 to 6 can only charge, and modes $3'$ to $6'$ can only discharge.

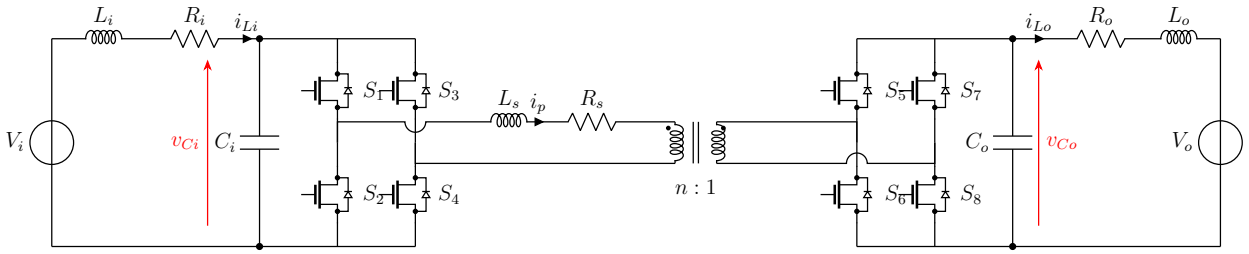


Figure 1: DAB circuit schematic with primary-referred AC-link inductance and resistance.

The analysis is based on a five-state model,

$$\mathbf{x}(t) = \begin{bmatrix} i_p & i_{Li} & i_{Lo} & v_{Ci} & v_{Co} \end{bmatrix}^T. \tag{1}$$

$$\mathbf{A}_k = \begin{bmatrix} -\dfrac{R_s}{L_s} & 0 & 0 & \dfrac{s_1^{(k)}}{L_s} & -\dfrac{n\, s_2^{(k)}}{L_s} \\ 0 & -\dfrac{R_i}{L_i} & 0 & -\dfrac{1}{L_i} & 0 \\ 0 & 0 & -\dfrac{R_o}{L_o} & 0 & -\dfrac{1}{L_o} \\ -\dfrac{s_1^{(k)}}{C_i} & \dfrac{1}{C_i} & 0 & 0 & 0 \\ \dfrac{n\, s_2^{(k)}}{C_o} & 0 & \dfrac{1}{C_o} & 0 & 0 \end{bmatrix}. \tag{2}$$

$$\mathbf{f} = \begin{bmatrix} 0 \\ \dfrac{V_i}{L_i} \\ \dfrac{V_o}{L_o} \\ 0 \\ 0 \end{bmatrix}. \tag{3}$$

The eight switching instants of one period are the angles at which either bridge changes state:

$$\theta \in \Big\{0,\ \varphi_1,\ \pi,\ \pi+\varphi_1,\ \varphi_3,\ \varphi_3+\varphi_2,\ \pi+\varphi_3,\ \pi+\varphi_3+\varphi_2\Big\} \pmod{2\pi},$$

with $\varphi_i = D_i\pi$. Sorting these into an ascending sequence $\theta_1 \le \theta_2 \le \cdots \le \theta_8$ gives the interval durations

$$t_k = \frac{\theta_{k+1} - \theta_k}{\omega}, \qquad k = 1, \ldots, 8, \qquad \theta_9 = \theta_1 + 2\pi,$$

where $\omega = 2\pi f_s$. Coincident edges yield $t_k = 0$ and therefore contribute the identity map, so the construction remains valid on mode boundaries and at unity duty.

The switching functions of the two bridges are evaluated at the interval midpoint $\bar{\theta}_k = \frac{1}{2}(\theta_k + \theta_{k+1})$:

$$s_1^{(k)} = \tfrac{1}{2}\Big[\operatorname{sgn}\big(\sin\bar{\theta}_k\big) - \operatorname{sgn}\big(\sin(\bar{\theta}_k - \varphi_1)\big)\Big], \tag{4}$$

$$s_2^{(k)} = \tfrac{1}{2}\Big[\operatorname{sgn}\big(\sin(\bar{\theta}_k - \varphi_3)\big) - \operatorname{sgn}\big(\sin(\bar{\theta}_k - \varphi_3 - \varphi_2)\big)\Big], \tag{5}$$

both taking values in $\{-1, 0, +1\}$.

Because the edges are sorted rather than enumerated, a single expression covers all twelve operating modes and both directions of power flow; the outer shift φ_3 may be negative, and the modulo operation places it correctly within the period. On each interval the dynamics are linear and time-invariant,

$$\dot{\mathbf{x}} = \mathbf{A}_k\mathbf{x} + \mathbf{f},$$

so the state at the end of the interval follows from the variation of constants formula,

$$\mathbf{x}(t + t_k) = e^{\mathbf{A}_k t_k}\,\mathbf{x}(t) + \left(\int_0^{t_k} e^{\mathbf{A}_k\tau}\,\mathrm{d}\tau\right)\mathbf{f} \;\equiv\; \mathbf{E}_k\,\mathbf{x}(t) + \boldsymbol{\psi}_k.$$

Evaluating $\boldsymbol{\psi}_k$ as $\mathbf{A}_k^{-1}(\mathbf{E}_k - \mathbf{I})\mathbf{f}$ is not possible in general, since $\mathbf{A}_k$ is singular for those switching states in which one or both bridges are clamped. Both blocks are instead obtained from a single augmented matrix exponential,

$$\exp\!\left(\begin{bmatrix}\mathbf{A}_k & \mathbf{f}\\ \mathbf{0} & 0\end{bmatrix} t_k\right) = \begin{bmatrix}\mathbf{E}_k & \boldsymbol{\psi}_k\\ \mathbf{0} & 1\end{bmatrix}, \tag{6}$$

which requires no matrix inverse and is exact to machine precision. Composing the eight interval maps in sequence gives the exact one-period map of the converter,

$$\mathbf{x}_{k+1} = \mathbf{G}_p(\mathbf{D})\,\mathbf{x}_k + \mathbf{H}_p(\mathbf{D}, \mathbf{V}), \tag{7}$$

with

$$\mathbf{G}_p = \mathbf{E}_8\mathbf{E}_7\cdots\mathbf{E}_1, \qquad \mathbf{H}_p = \boldsymbol{\psi}_8 + \mathbf{E}_8\boldsymbol{\psi}_7 + \mathbf{E}_8\mathbf{E}_7\boldsymbol{\psi}_6 + \cdots + \mathbf{E}_8\cdots\mathbf{E}_2\,\boldsymbol{\psi}_1. \tag{8}$$

Both are accumulated in a single pass over the intervals,

$$\mathbf{G}_p \leftarrow \mathbf{E}_k\mathbf{G}_p, \qquad \mathbf{H}_p \leftarrow \mathbf{E}_k\mathbf{H}_p + \boldsymbol{\psi}_k,$$

initialised with $\mathbf{G}_p = \mathbf{I}$ and $\mathbf{H}_p = \mathbf{0}$.

Since every interval map is exact, (7) carries no integration error and reproduces the switching converter to machine precision. The matrix $\mathbf{G}_p$ is the monodromy matrix of the periodic solution, and its eigenvalues are the Floquet multipliers. The steady-state operating point is the fixed point of (7). Imposing periodicity, $\mathbf{x}_{k+1} = \mathbf{x}_k = \mathbf{x}^\star$, gives

$$\mathbf{x}^\star = (\mathbf{I} - \mathbf{G}_p)^{-1}\,\mathbf{H}_p, \tag{9}$$

which is obtained in closed form, without iteration or transient simulation. The inverse exists whenever no Floquet multiplier lies at $+1$, guaranteed here by the positive resistance present in every loop.

Because the one-period map is affine in the state, the state Jacobian is the monodromy matrix itself:

$$\mathbf{A}_d = \frac{\partial \mathbf{x}_{k+1}}{\partial \mathbf{x}_k} = \mathbf{G}_p \qquad \text{exactly.} \tag{10}$$

No symbolic differentiation and no finite difference is required for $\mathbf{A}_d$; the same quantity that propagates the state also linearises it. The input Jacobians $\mathbf{B}$, with respect to D_1, D_2, D_3, and $\mathbf{C}$, with respect to V_i, V_o, are, as in Krismer and Kolar [1], the derivatives of the state at the end of the period with respect to the inputs; here they are evaluated by central differences of the exact period map (7) about $\mathbf{x}^\star$.

Collecting the three modulation variables and the two source voltages into a single input vector $\mathbf{u} = [\, D_1 \;\; D_2 \;\; D_3 \;\; V_i \;\; V_o \,]^T$, the linearised model is

$$\delta \mathbf{x}_{k+1} = \mathbf{A}_d \, \delta \mathbf{x}_k + \begin{bmatrix} \mathbf{B} & \mathbf{C} \end{bmatrix} \delta \mathbf{u}_k, \qquad \delta \mathbf{y}_k = \mathbf{E}_y \, \delta \mathbf{x}_k, \tag{11}$$

with transfer-function matrix

$$\mathbf{G}(z) = \mathbf{E}_y \, (z\mathbf{I} - \mathbf{A}_d)^{-1} \begin{bmatrix} \mathbf{B} & \mathbf{C} \end{bmatrix} \in \mathbb{C}^{1 \times 5}. \tag{12}$$

where $\mathbf{E}_y = [\, 0 \;\; 0 \;\; 1 \;\; 0 \;\; 0 \,]$ selects the output-branch current i_{Lo}. The first three columns are control paths and the last two are disturbance paths, since the port voltages are imposed by the supply and the battery rather than commanded. The entry used for the single-loop design is the third, $G_{i_{Lo},D_3}(z)$, from D_3 to i_{Lo}, whose value at $z = 1$ is the control gain

$$K_{\mathrm{dc}} = \mathbf{E}_{y,1} \, (\mathbf{I} - \mathbf{A}_d)^{-1} \, \mathbf{B}_{:,3} = \left. \frac{\partial i_{Lo}}{\partial D_3} \right|_{\mathbf{x}^\star}. \tag{13}$$

Sampling the output current, computing the correction and updating the pulse-width modulator each cost time. Writing N for the number of whole cycles between the sampling instant and the instant at which the new duty takes effect, the plant the controller actually sees is

$$G_{\mathrm{plant}}(z) = G_{i_{Lo},D_3}(z) \, z^{-N}, \tag{14}$$

with $N = 1$ in this work. The delay leaves the DC gain unchanged and subtracts a phase $-N\omega T_s$, growing linearly with frequency.

[1] F. Krismer and J. W. Kolar, “Accurate small-signal model for an automotive bidirectional dual active bridge converter,” IEEE, 2008.